

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

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Course Code: ITA0609

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Course Title: Machine Learning

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CO3 ASSESSMENT TOOL 2 – CASE STUDY

Case Study 1 : A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using Bayesian Belief Networks and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

Title

Bayesian Belief Network-Based Smart Healthcare System for Real-Time Anomaly Detection

1. Problem

Patients in hospitals or remote healthcare environments need continuous monitoring of vital signs such as **heart rate, SpO₂, blood pressure, and body temperature**. Wearable sensors may produce noisy or incomplete readings, making it difficult to determine whether an abnormal value indicates a real medical condition.

2. Objective

To develop a probabilistic healthcare system that uses **Bayesian Belief Networks (BBN)** to identify abnormal patient conditions while handling uncertainty and sensor noise.

3. Case Scenario

A patient wears a smart monitoring device that continuously measures:

- Heart Rate
- SpO₂
- Blood Pressure
- Body Temperature
- Respiratory Rate

The system learns the probabilistic relationship between these parameters and health conditions.

For example:

Low SpO₂ → Increased Heart Rate → Increased Probability of Respiratory Problem

The Bayesian network calculates the probability of an abnormal condition instead of making a decision based on only one sensor reading.

4. Bayesian Model

Example dependencies:

SpO₂ → Respiratory Condition

Heart Rate → Respiratory Condition

Temperature → Infection

Blood Pressure → Cardiovascular Condition

The system combines these probabilities to estimate the patient's overall health state.

5. Sample Input

Heart Rate	SpO ₂	Temperature	Blood Pressure
110 bpm	88%	37.2°C	145/90

6. Sample Output

Predicted condition: Possible respiratory/cardiovascular abnormality

Probability of abnormal condition: High

Alert: Immediate monitoring required

7. Handling Sensor Noise

Sensor readings can be inaccurate because of **movement, poor sensor contact, battery problems, or environmental interference.**

Instead of treating every measurement as perfectly accurate, the Bayesian model assigns probabilities to observations.

For example:

Measured SpO₂ = 88%

Probability that actual SpO₂ is below normal = 0.92

Thus, uncertainty is explicitly incorporated into the decision.

8. Performance Evaluation

Important metrics:

- Detection accuracy

- False alarm rate
- Sensitivity
- Specificity
- Response time

A successful system should detect genuine abnormalities while minimizing unnecessary alerts.

9. Challenges

- Sensor noise and missing readings
- Real-time processing requirements
- Patient-to-patient variation
- Limited wearable battery power
- Data privacy and security
- Need for reliable medical validation

10. Conclusion

The Bayesian healthcare system can provide **probabilistic real-time anomaly detection** by combining multiple physiological parameters. It is particularly useful when sensor data is uncertain or incomplete, helping healthcare professionals identify potentially dangerous conditions earlier.

Case Study 2: A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply Naïve Bayes classifiers, EM algorithm, and Bayesian learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments

Title

Probabilistic Learning-Based Network Intrusion Detection Using Naïve Bayes, EM and Bayesian Learning

1. Problem

Modern computer networks continuously generate large amounts of traffic. Identifying malicious activities such as **DoS attacks, port scanning, malware communication, and unauthorized access** is difficult because network behavior changes over time.

2. Objective

To develop a probabilistic intrusion detection system that identifies **normal and abnormal network behavior**, handles incomplete data, and reduces false alarms.

3. Case Scenario

A company monitors network traffic using parameters such as:

- Protocol
- Source IP behavior
- Destination port
- Packet rate
- Connection duration
- Number of failed connections
- Bytes transferred

The system learns the probability of a connection being **normal or an intrusion**.

4. Naïve Bayes Model

For a network connection:

$P(\text{Intrusion} \mid \text{Network Features})$

is calculated using the probabilities of the observed features.

Example:

- High packet rate
- Large number of failed connections
- Unusual port
- Short connection duration

These characteristics may increase the probability of an intrusion.

5. Sample Input

Packet Rate	Failed Connections	Port	Duration
950 packets/s	85	23	2 sec

6. Sample Output

Prediction: Intrusion

Intrusion Probability: 96%

Alert: Suspicious network activity detected

7. Role of EM Algorithm

The **Expectation-Maximization (EM)** algorithm is useful when some network information is missing or when the data contains hidden patterns.

It works in two major steps:

E-step: Estimate hidden variables/probabilities.

M-step: Update model parameters using those estimates.

This process is repeated until the model converges.

For example, the system may discover hidden groups of network behavior such as:

- Normal users
- Automated traffic
- Suspicious scanning
- Attack traffic

8. Bayesian Learning

Bayesian learning updates the model when new network data becomes available.

For example:

Initial belief → New network evidence → Updated probability

This allows the intrusion detection system to adapt to changing attack patterns.

9. Handling Incomplete Data

If one feature is missing, probabilistic methods can estimate its contribution using available information rather than immediately discarding the entire network record.

10. Performance Evaluation

The system can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- False Positive Rate
- False Negative Rate

A major objective is to **reduce false alarms** while maintaining high intrusion detection accuracy.

11. Challenges

- Rapidly changing attack patterns
- Large network traffic volume
- Zero-day attacks
- Imbalanced datasets
- Missing network features
- High computational requirements
- False positives from legitimate unusual behavior

12. Conclusion

Combining **Naïve Bayes, EM, and Bayesian learning** provides a flexible probabilistic framework for intrusion detection. Naïve Bayes provides fast classification, EM discovers hidden patterns and handles incomplete information, while Bayesian learning allows continuous model updating.

Case Study 3: An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using Bayesian inference and probability learning, design a predictive model. Analyze how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.

Title

Bayesian Inference-Based Crop Yield Prediction Under Uncertain Environmental Conditions

1. Problem

Crop yield depends on uncertain factors such as **rainfall, temperature, soil quality, irrigation, and seasonal conditions**. Traditional prediction methods may not adequately represent this uncertainty.

2. Objective

To develop a Bayesian predictive model that estimates crop yield using environmental and agricultural parameters while incorporating prior agricultural knowledge.

3. Case Scenario

Consider a farmer growing **rice**.

The system collects:

- Rainfall
- Average temperature
- Soil quality
- Irrigation availability
- Previous crop yield
- Fertilizer usage
- Season

The Bayesian model estimates the expected crop yield.

4. Bayesian Model

Possible relationships:

Rainfall → Soil Moisture → Crop Yield

Temperature → Crop Growth → Crop Yield

Soil Quality → Crop Growth → Crop Yield

Season → Rainfall/Temperature → Crop Yield

The model combines these dependencies to predict yield.

5. Sample Input

Parameter	Value
Rainfall	850 mm
Temperature	28°C
Soil Quality	Good
Irrigation	Available
Fertilizer	Recommended
Season	Kharif

6. Sample Output

Predicted Crop Yield: 4.2 tonnes/hectare

Yield Category: High

Prediction Confidence: 87%

7. Prior Agricultural Knowledge

Bayesian inference allows existing agricultural knowledge to be used as a **prior probability**.

For example, historical agricultural records may indicate:

Good-quality soil + adequate rainfall → High probability of good rice yield.

When new seasonal data becomes available, the model updates this belief.

Prior Knowledge + New Evidence → Posterior Prediction

8. Seasonal Variation

Weather conditions differ between seasons.

For example:

- High rainfall may benefit crops during one season.
- Excessive rainfall may cause flooding during another.
- High temperature may reduce yield during a drought period.

The Bayesian model can update predictions according to the observed seasonal conditions.

9. Handling Uncertainty

Instead of giving only:

Predicted yield = 4 tonnes/hectare

the system can provide:

Expected yield = 4 tonnes/hectare

Possible range = 3.5–4.5 tonnes/hectare

Confidence = 85%

This gives farmers more useful information for decision-making.

10. Decision-Making

The prediction can help farmers decide:

- Whether irrigation is required
- Amount of fertilizer needed
- Suitable crop selection
- Expected production
- Harvest planning
- Risk management

At the policy level, governments can use predictions for:

- Food supply planning
- Agricultural subsidies
- Crop insurance
- Water-resource planning
- Disaster preparedness

11. Performance Evaluation

The model can be evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Prediction accuracy
- Confidence/calibration

- Seasonal prediction reliability

12. Challenges

- Unpredictable weather
- Limited historical data
- Regional differences in soil
- Climate change
- Missing agricultural records
- Changes in farming practices

13. Conclusion

Bayesian inference provides a reliable approach for **crop yield prediction under uncertainty**. By combining historical agricultural knowledge with current environmental observations, the model can provide probabilistic predictions that support both **farmer-level decisions and agricultural policy planning**.

Quick Comparison of All 3

Case Study	Main Application	Bayesian Technique	Main Uncertainty
Healthcare	Patient anomaly detection	Bayesian Belief Network	Sensor noise & missing vital signs
Cybersecurity	Intrusion detection	Naïve Bayes + EM + Bayesian Learning	Incomplete data & changing attacks
Agriculture	Crop yield prediction	Bayesian Inference	Weather & environmental variation